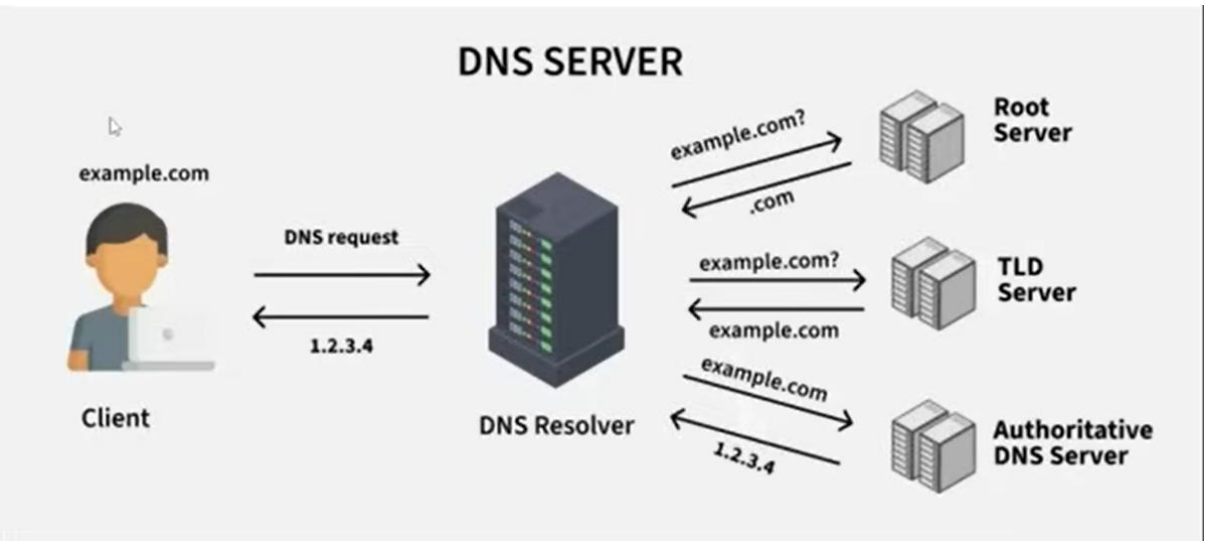
DNS SPOOFING OR DNS CACHE POISONING

DNS spoofing, or DNS cache poisoning, is a cyberattack that corrupts a Domain Name System (DNS) resolver's cache with false information. By altering the IP address associated with a website, it redirects user traffic to malicious or fraudulent sites, aiming to steal data (phishing) or install malware.



DNS cache spoofing, also known as **DNS cache poisoning**, is a type of cyberattack that involves inserting false Domain Name System (DNS) data into a DNS resolver's cache. This manipulation causes the name server to return an incorrect IP address, which redirects users attempting to visit a legitimate website to a malicious one without their knowledge.

How DNS Cache Spoofing Works

The Domain Name System acts as the internet's phonebook, translating human-readable domain names (like example.com) into machine-readable IP addresses. To speed up this process, DNS servers and user devices temporarily store these translations in a cache.

In a DNS spoofing attack:

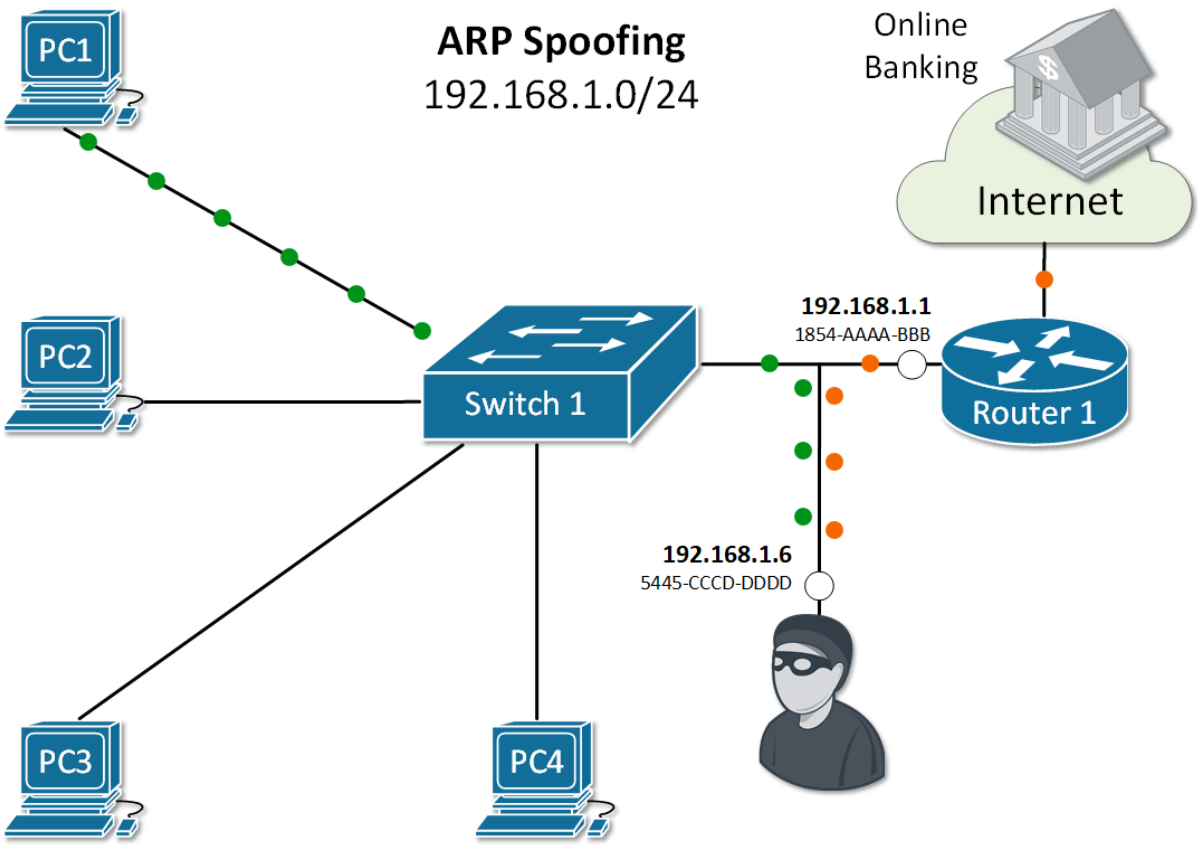
1. **Reconnaissance:** The attacker gathers information about the target DNS server's vulnerabilities, software, and query patterns.
2. **Manipulation:** The attacker exploits a vulnerability in the DNS protocol, such as its use of the unencrypted User Datagram Protocol (UDP), to send a forged DNS response to a recursive DNS resolver. This response contains a malicious IP address but appears to be from a legitimate source.
3. **Poisoning:** If the fake response is accepted and cached by the resolver (which happens before the legitimate response arrives), the cache is "poisoned".
4. **Redirection:** Any subsequent user who requests the legitimate website name from that poisoned DNS server is sent to the attacker's server instead, where a fraudulent website is hosted. The entry remains in the cache until its time-to-live (TTL) expires or is manually removed.

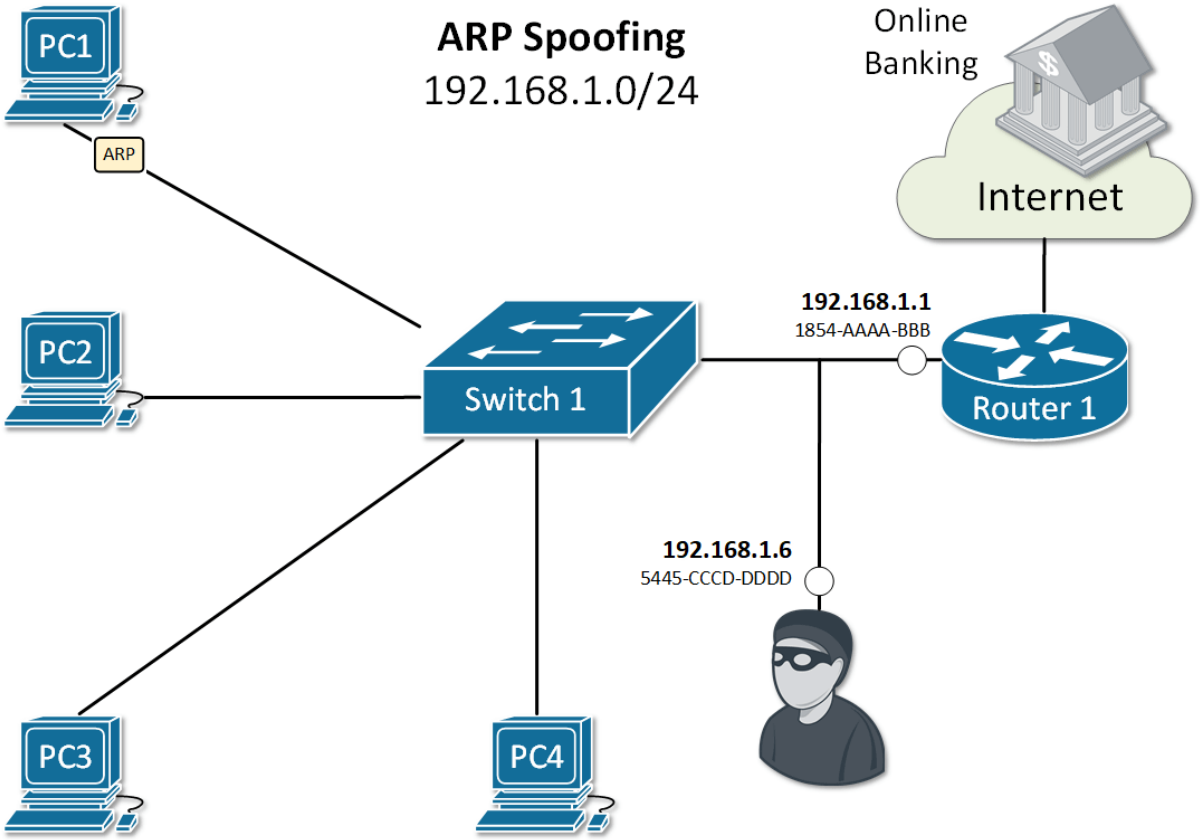
A Man-in-the-Middle (MITM) attack is a cyber threat where an attacker secretly intercepts and relays communications between two parties who believe they are communicating directly. By inserting themselves into the network flow, criminals steal credentials, financial data, or modify messages in real-time.

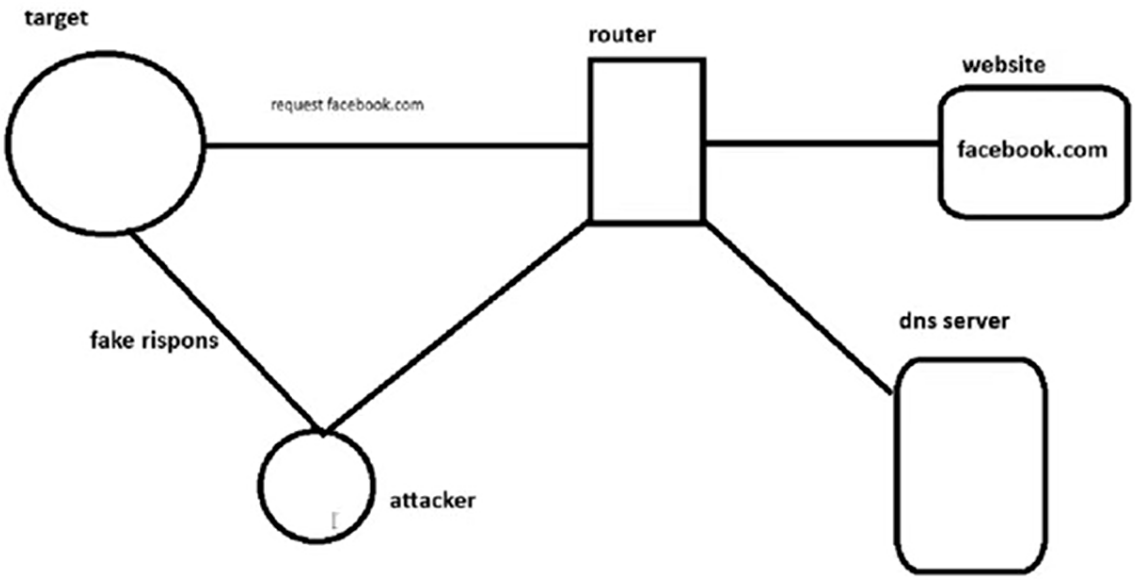
ARP spoofing

When a rouge host in the local area network sends false ARP messages, it can associate its MAC address with the IP address of the default gateway or any other IP in the LAN. It then can start receiving data intended for another host in the network. There are there main attacks that utilize this spoofing technique:

- **Main-in-the-Middle:** An intruder intercepts ARP messages from legitimate users and modifies them to redirect the data.
- **Session Hijacking:** A hacker tries to use ARP spoofing to steal a cookie, token, or session ID so he can gain unauthorized access to private data. Pretty common in public WiFi networks at busy bus stations and airports.
- **Denial of Service (DoS)**: An attacker tries to make a host or network resource unavailable to its intended users. Typically in the case of ARP spoofing, it is done by stealing legitimate data intended to the particular host or flooding the host with a high number of false or malformed ARP messages.

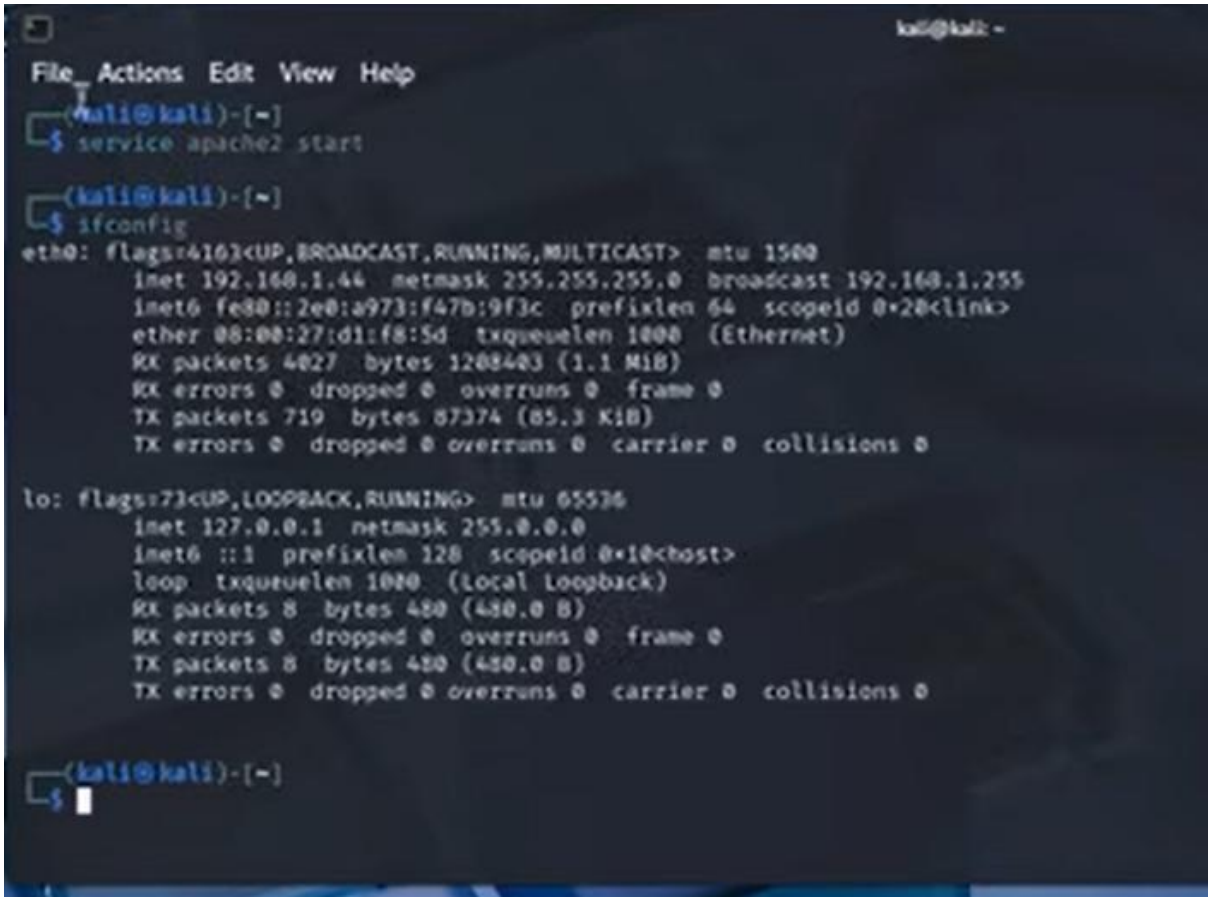



ARP Spoofing
192.168.1.0/24
Online
Banking
Internet
PC1
ARP
PC2
Switch 1
192.168.1.1
1854-AAAA-BBB
Router 1
192.168.1.6
5445-CCCD-DDDD
PC3
PC4

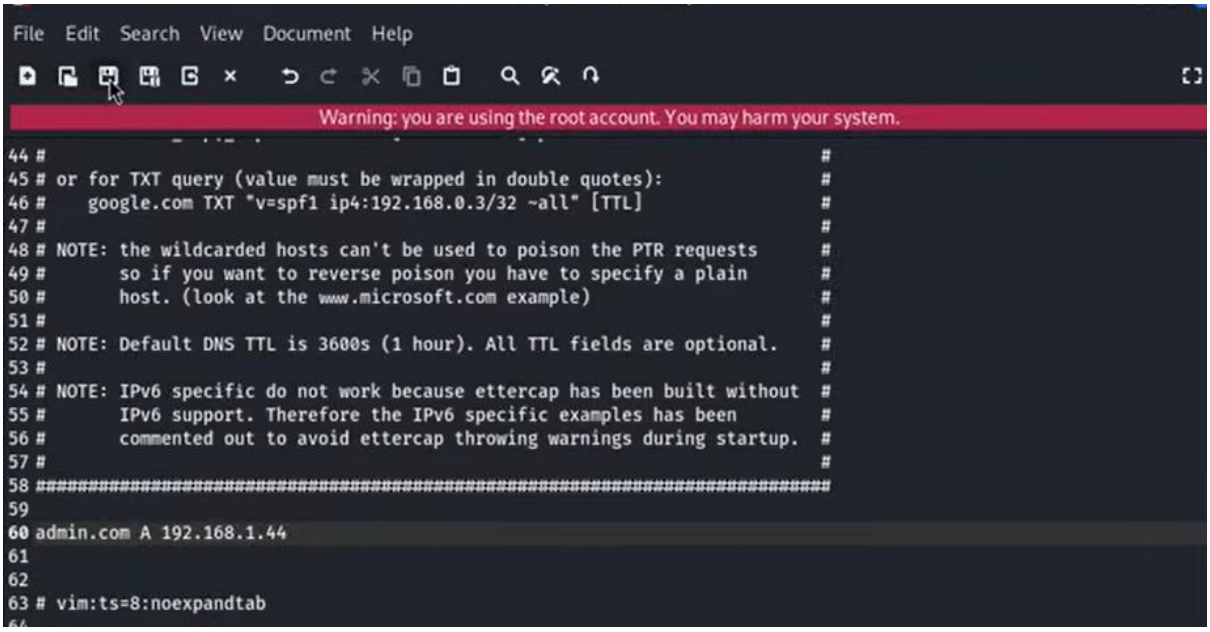

target
router
website
request facebook.com
facebook.com
fake rispons
dns server
attacker

EXERCISE

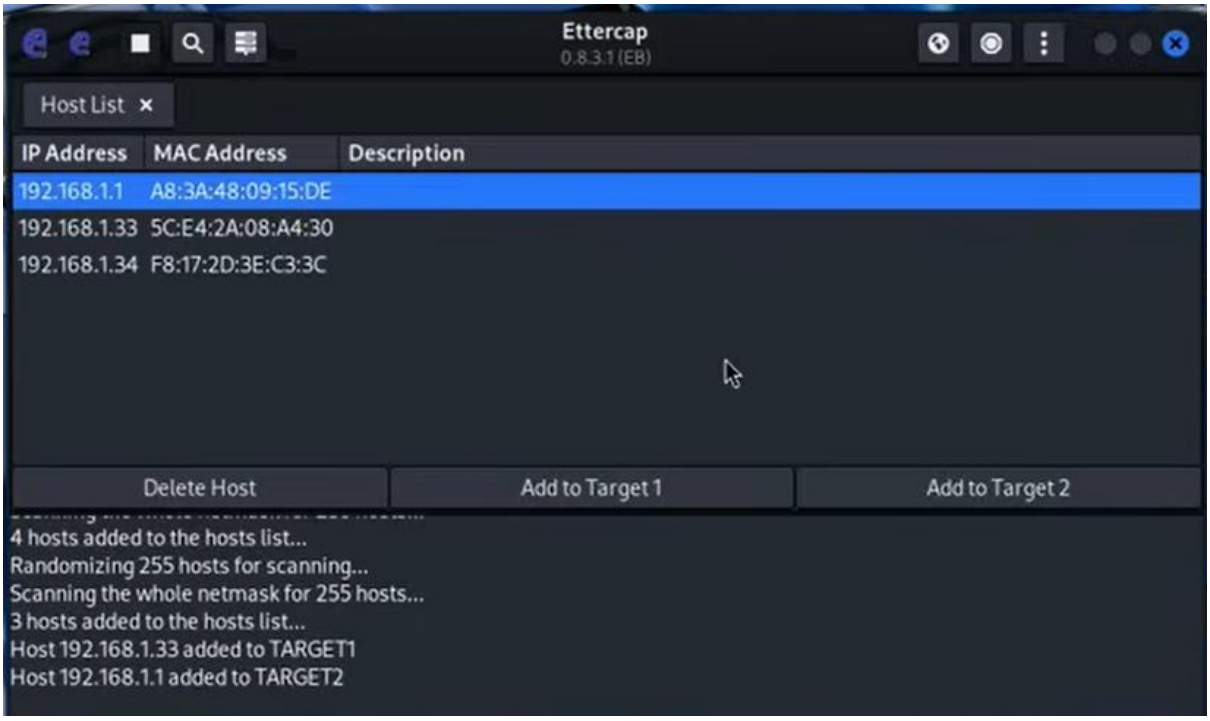
1. Create your own website
2. Start Apache Server
   $ service apache2 start
3. Check the IP (copy the IP address)
   $ ifconfig



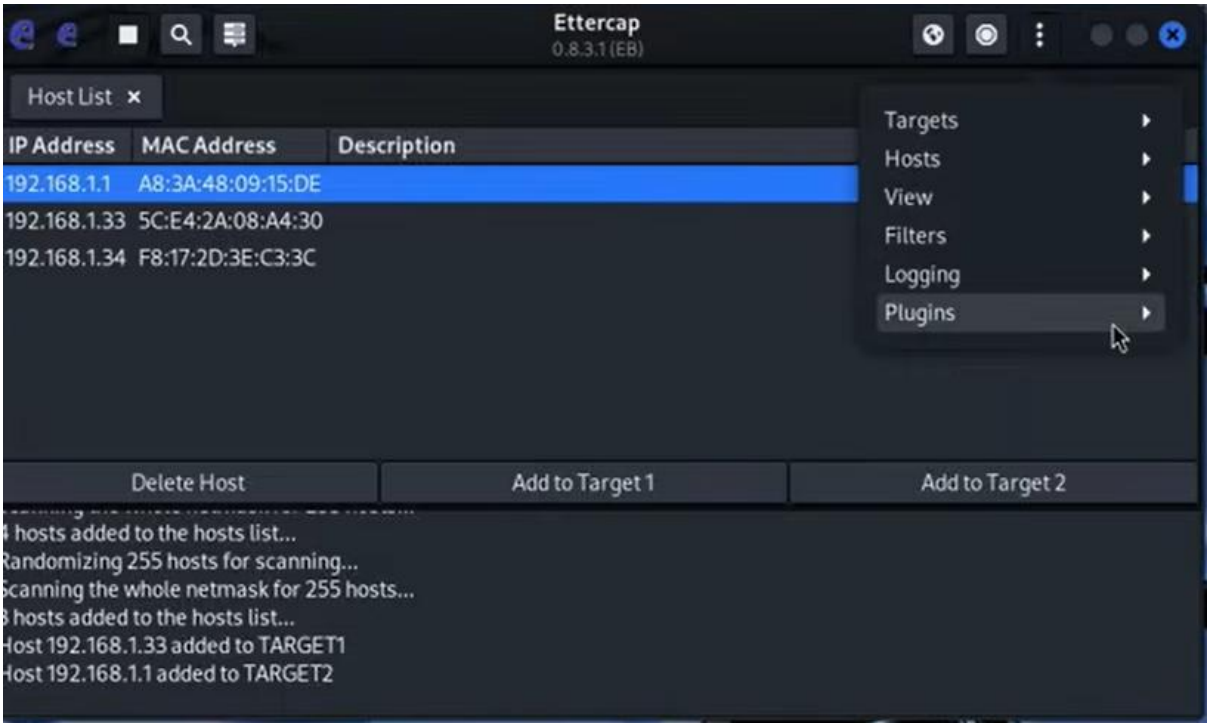
4. Make the following entry into ettercap file system
   Filesystem > etc > Ettercap > etter.dns
   Original domain name A new IP Address
   # Xyz.com A 192.123.3.5



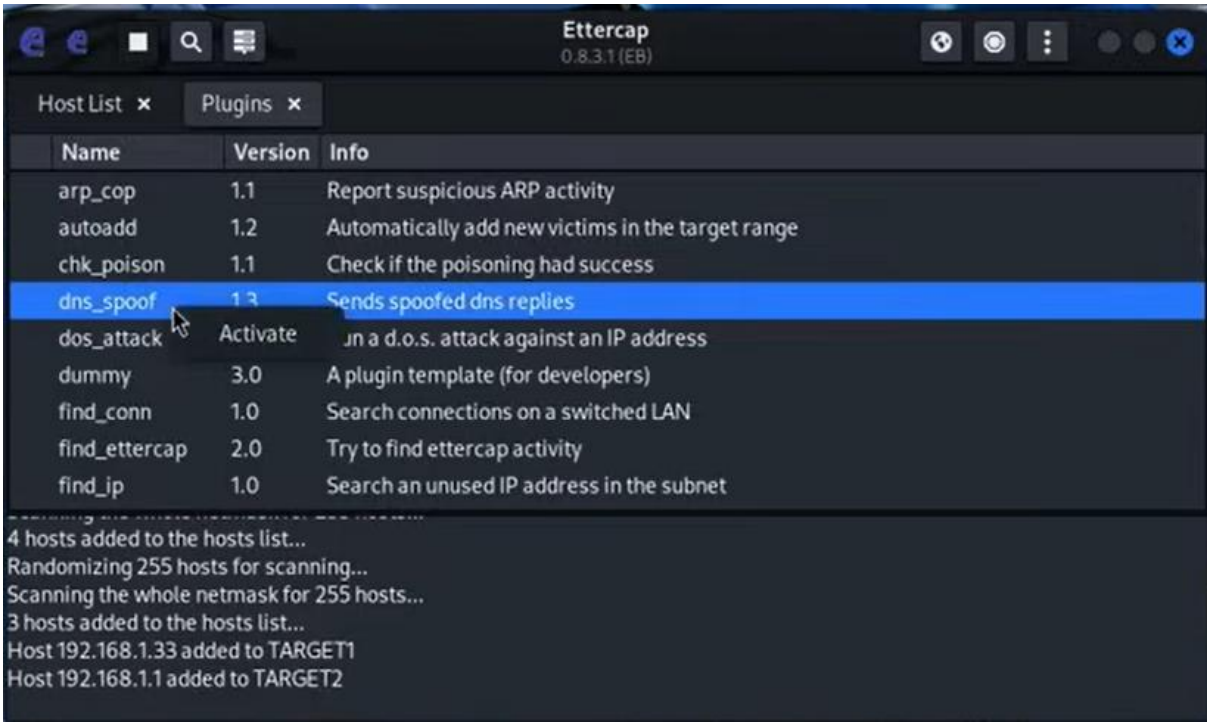
5. Save the file etter.dns
6. Open Ettercap Tool
7. Scan for hosts connected
8. Now choose the Target 1(Host System) and Target 2(Router)



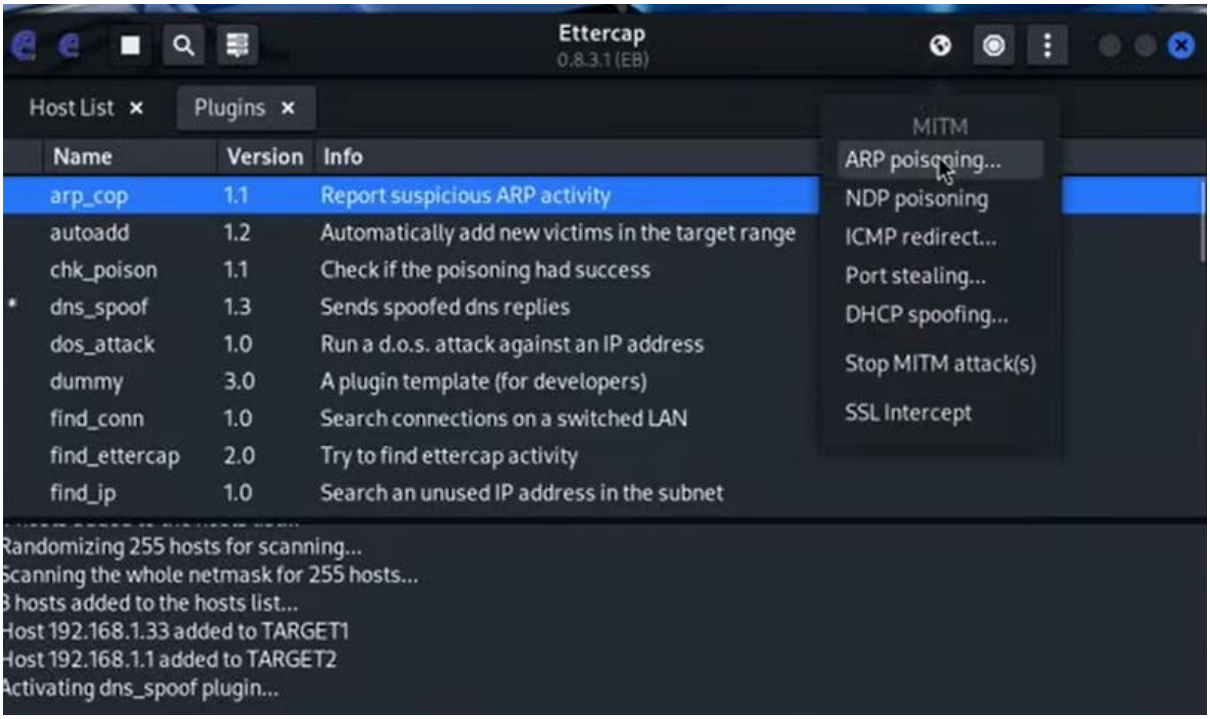
9. Now go to Plugins

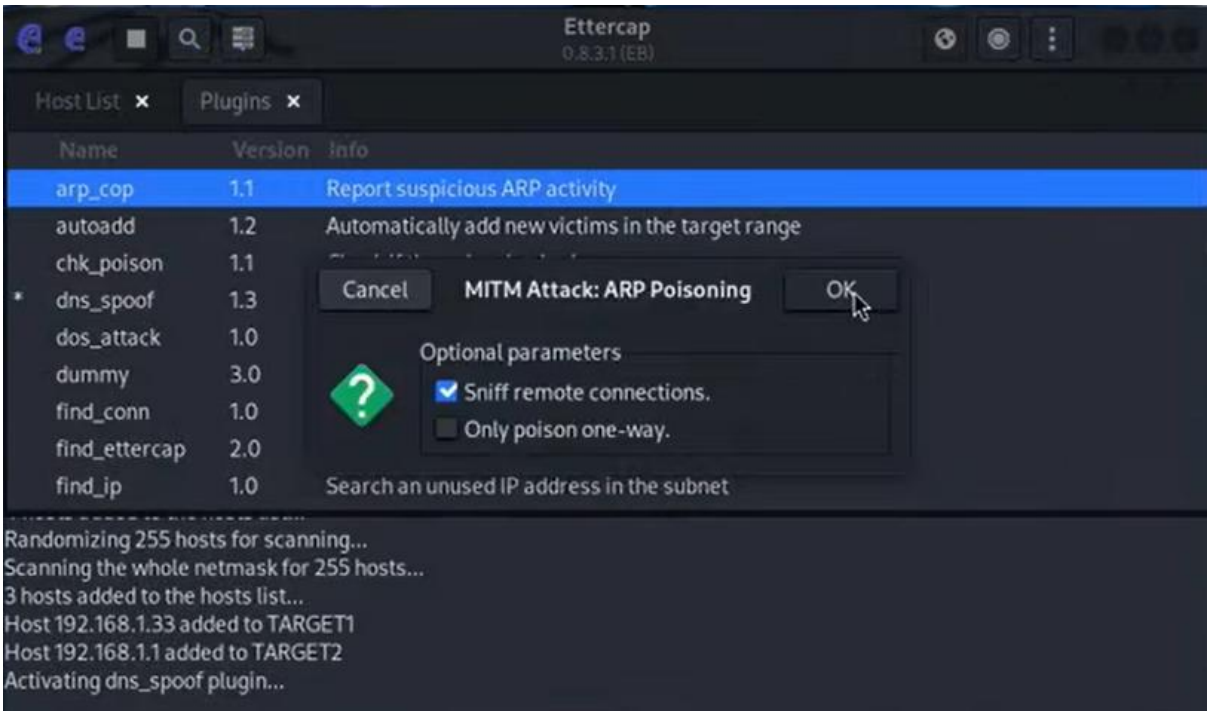


10.Go to Manage Plugins > DNS SPOOF > ACTIVATE



11. MITM > ARP POISIONING





12. In windows, open xyz.com